

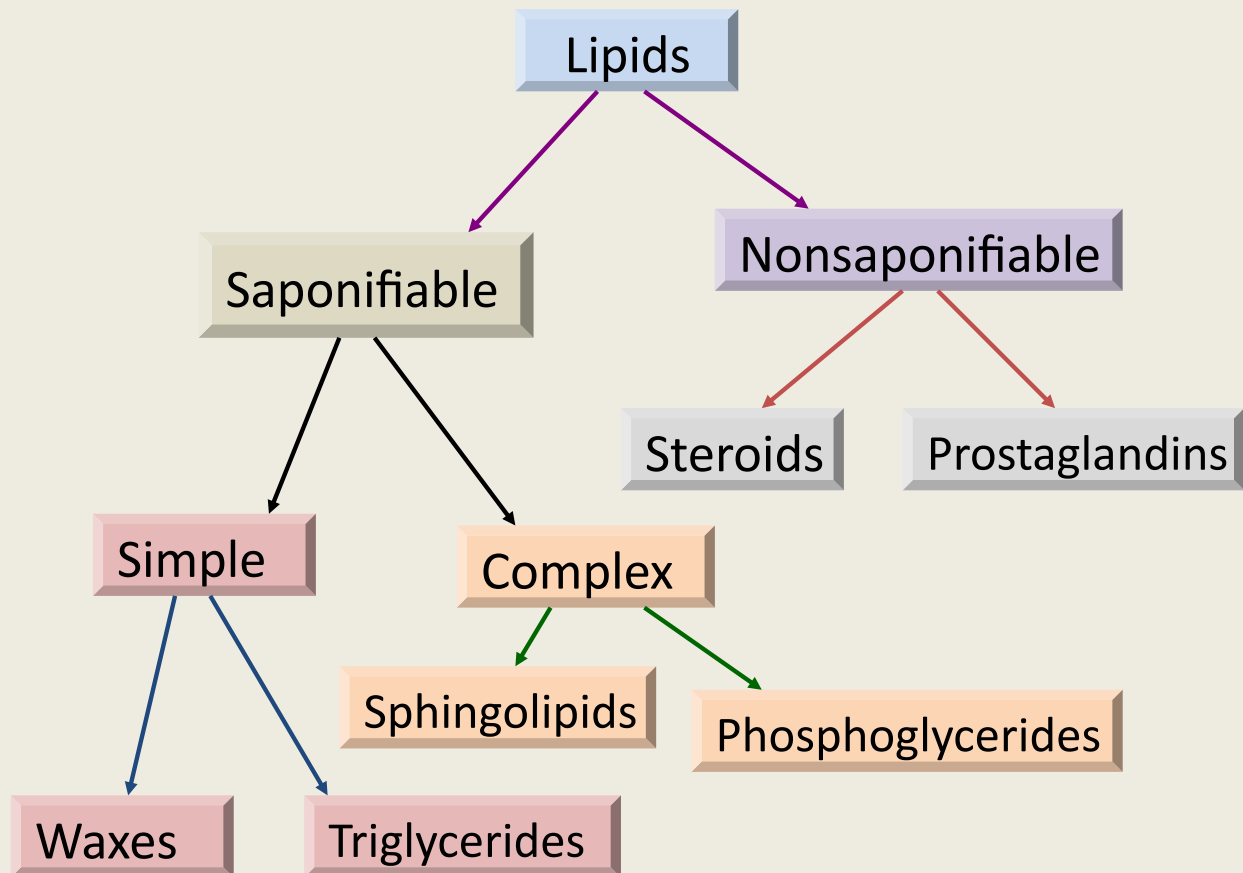
Lipids

Chapter 18

Lipids

- Biological compounds that are soluble only in nonpolar solvents.
- Saponifiable lipids contain an ester that can undergo basic hydrolysis.
 - Simple lipids contain one or more fatty acids and an alcohol.
 - Triglycerides and waxes
 - Complex lipids are ester-containing lipids with more than two types of components: an alcohol, fatty acids, and other components.
 - Phospholipids and sphingolipids
- Nonsaponifiable lipids do not contain an ester and cannot be hydrolyzed.
 - Steroids and prostaglandins

Lipid Classification

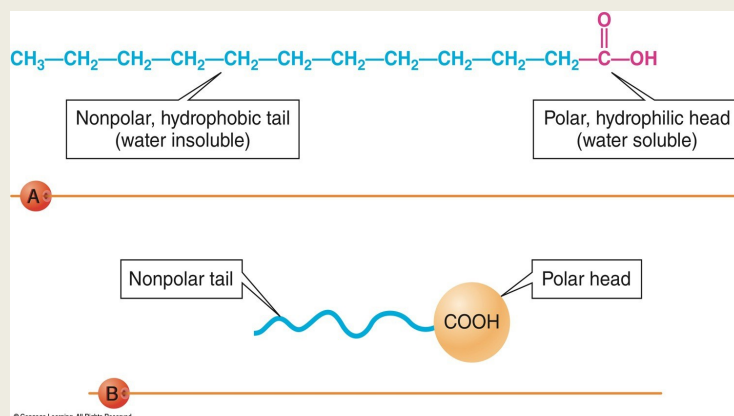


Functions of Lipids

- Energy source and reserve in plants and animals
- Essential components of biological membranes
- Thermal insulation in polar and hibernating animals
- Source of metabolic water in animals such as camels
- In plants provide energy and biosynthetic precursors during seed germination

Fatty Acids

- Are the building blocks of many lipids.
- Are long chain carboxylic acids.
- Have long nonpolar tails responsible for fatty/oily characteristics.
- The carboxyl group is very hydrophilic under conditions of physiological pH (exists as COO^-).



Examples of Fatty Acids

Name	Formula
Stearic acid	$\text{CH}_3(\text{CH}_2)_{16}-\text{COOH}$
Linoleic acid ^a	$\text{CH}_3(\text{CH}_2)_4(\text{CH}=\text{CHCH}_2)_2(\text{CH}_2)_6-\text{COOH}$
Linolenic acid ^b	$\text{CH}_3\text{CH}_2(\text{CH}=\text{CHCH}_2)_3(\text{CH}_2)_6-\text{COOH}$
Oleic acid	$\text{CH}_3(\text{CH}_2)_7\text{CH}=\text{CH}(\text{CH}_2)_7-\text{COOH}$

Fatty Acids

- Fatty acids are:
 - Usually straight chains (no branching).
 - Usually from C_{10} to C_{20} .
 - Usually an even number of carbons.
 - Either saturated (no $C=C$ bonds) or unsaturated (has $C=C$ bonds, usually in the *cis* configuration).

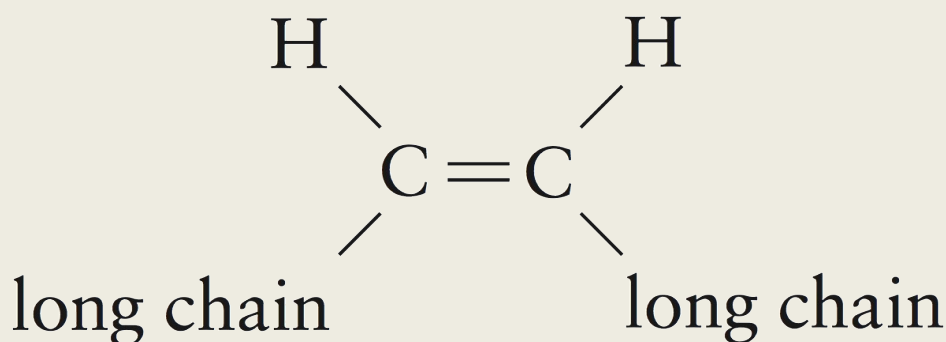


TABLE 18.1 Some Important Fatty Acids

Compound Type and Number of Carbons	Name	Formula	Melting Point (°C)	Common Sources
<i>Saturated</i>				
14	Myristic acid	$CH_3(CH_2)_{12}-COOH$	54	Butterfat, coconut oil, nutmeg oil
16	Palmitic acid	$CH_3(CH_2)_{14}-COOH$	63	Lard, beef fat, butterfat, cottonseed oil
18	Stearic acid	$CH_3(CH_2)_{16}-COOH$	70	Lard, beef fat, butterfat, cottonseed oil
20	Arachidic acid	$CH_3(CH_2)_{18}-COOH$	76	Peanut oil
<i>Monounsaturated</i>				
16	Palmitoleic acid	$CH_3(CH_2)_5CH=CH(CH_2)_7-COOH$	-1	Cod liver oil, butterfat
18	Oleic acid	$CH_3(CH_2)_7CH=CH(CH_2)_7-COOH$	13	Lard, beef fat, olive oil, peanut oil
<i>Polyunsaturated</i>				
18	Linoleic acid ^a	$CH_3(CH_2)_4(CH=CHCH_2)_2(CH_2)_6-COOH$	-5	Cottonseed oil, soybean oil, corn oil, linseed oil
18	Linolenic acid ^b	$CH_3CH_2(CH=CHCH_2)_3(CH_2)_6-COOH$	-11	Linseed oil, corn oil
20	Arachidonic acid ^a	$CH_3(CH_2)_4(CH=CHCH_2)_4(CH_2)_2-COOH$	-50	Corn oil, linseed oil, animal tissues
20	Eicosapentaenoic acid ^b	$CH_3CH_2(CH=CHCH_2)_5(CH_2)_2-COOH$		Fish oil, seafoods
22	Docosahexaenoic acid ^b	$CH_3CH_2(CH=CHCH_2)_6CH_2-COOH$		Fish oil, seafoods

^aOmega-6 fatty acid.

^bOmega-3 fatty acid.

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Fatty Acids

- Essential fatty acids are those needed by the body, but not synthesized within the body in adequate amounts.
- For humans, linoleic acid and linolenic acid are essential, easily obtainable from plant and fish oils.



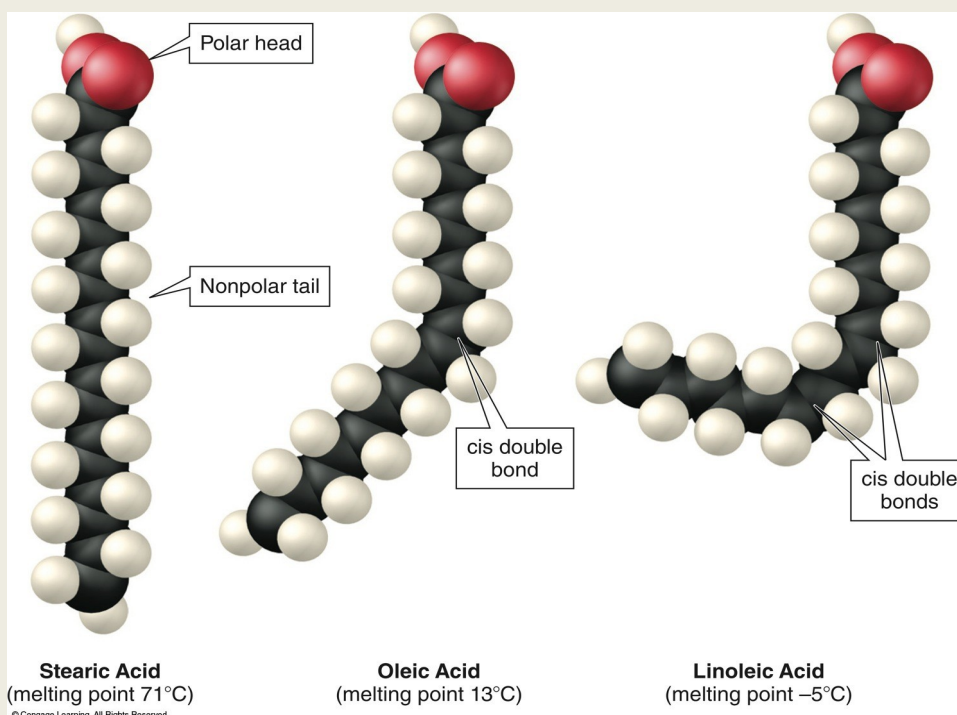
linoleic acid



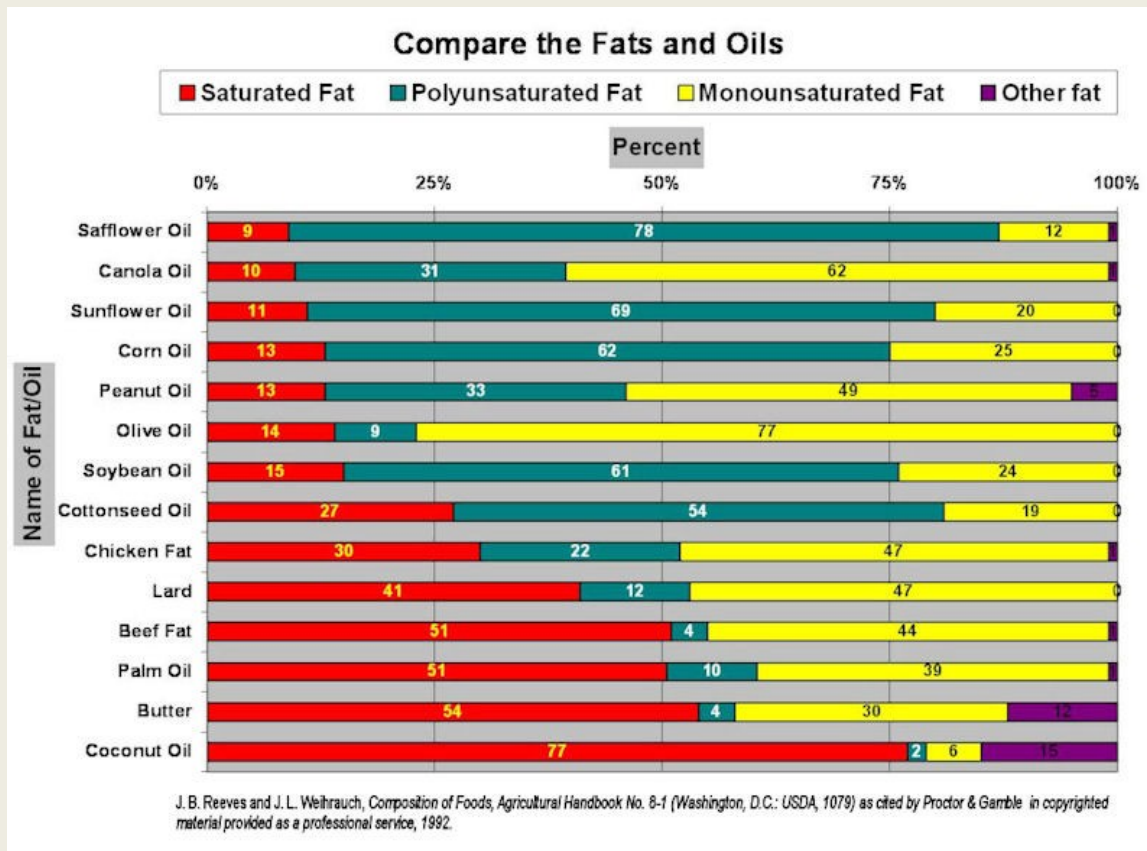
linolenic acid

Fatty Acids

- Cis carbon-carbon double bonds causes “kinking” of the carbon chain.



Sources of Fatty Acids



Comparison between saturated and unsaturated fatty acids

Properties of Unsaturated Fatty Acids

- Contain one or more double C=C bonds
- Nonlinear chains do not allow molecules to pack closely
- Few interactions between chains
- Low melting points
- Liquids at room temperature



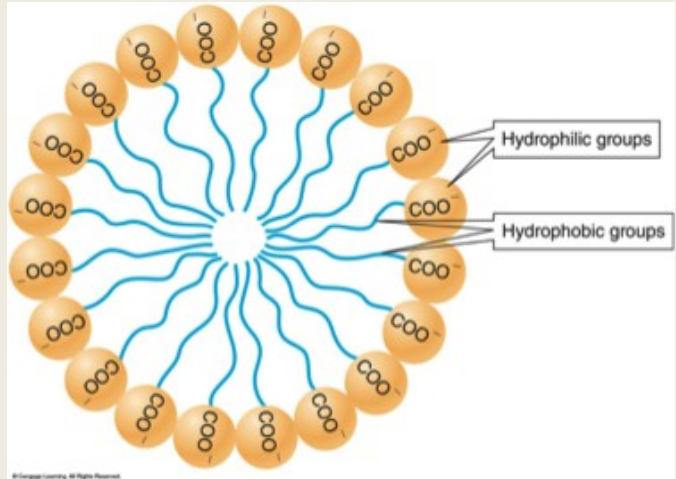
Properties of Saturated Fatty Acids

- Contain only single C-C bonds
- Closely packed
- Strong attractions between chains
- High melting points
- Solids at room temperature



Fatty Acids

- In water, fatty acids will form spherical clusters called micelles.
- Micelles are important for biological functions, like the transport of insoluble lipids in the blood.

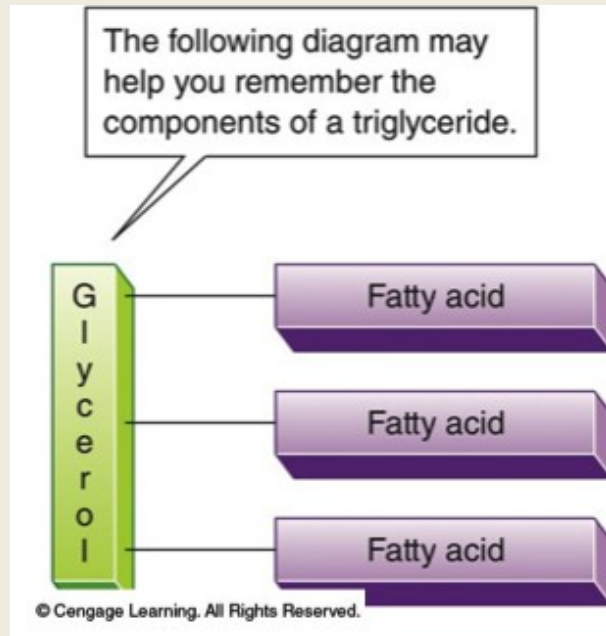


Fats and Oils

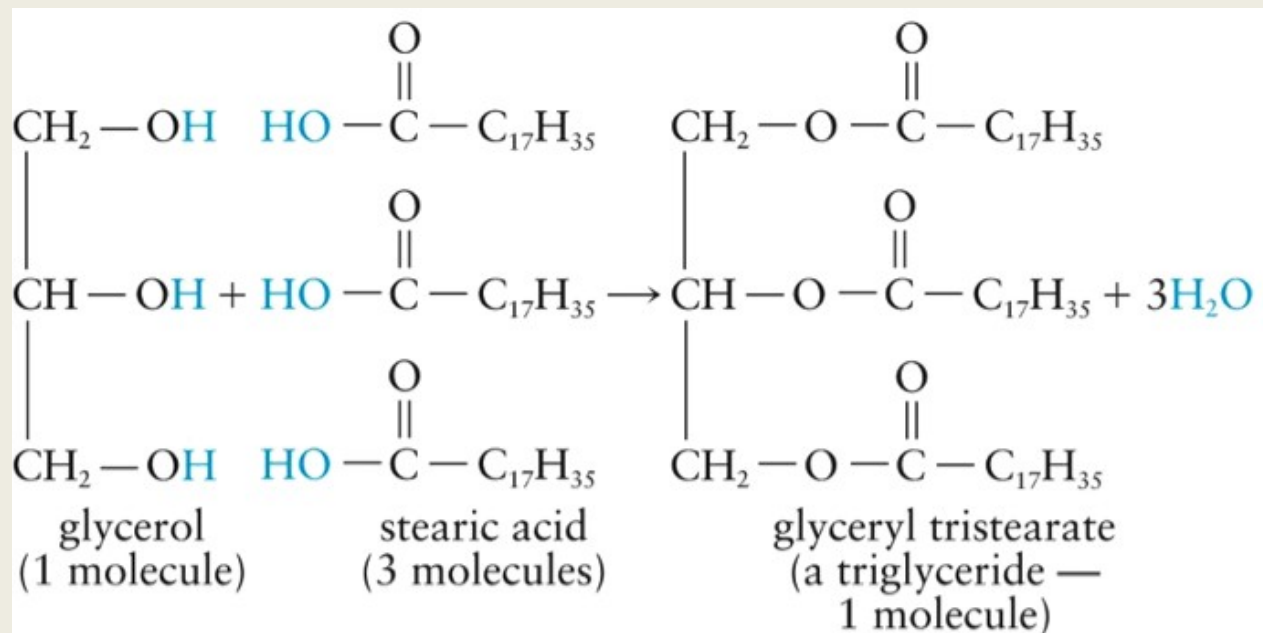
- Fats are:
 - Usually from animal sources.
 - Solids at room temperature.
 - Usually composed to a high degree of saturated fatty acids.
- Oils are:
 - Usually from plant and fish sources.
 - Liquids at room temperature.
 - Usually composed of more unsaturated fatty acids than saturated fatty acids.

Structure of Fats and Oils

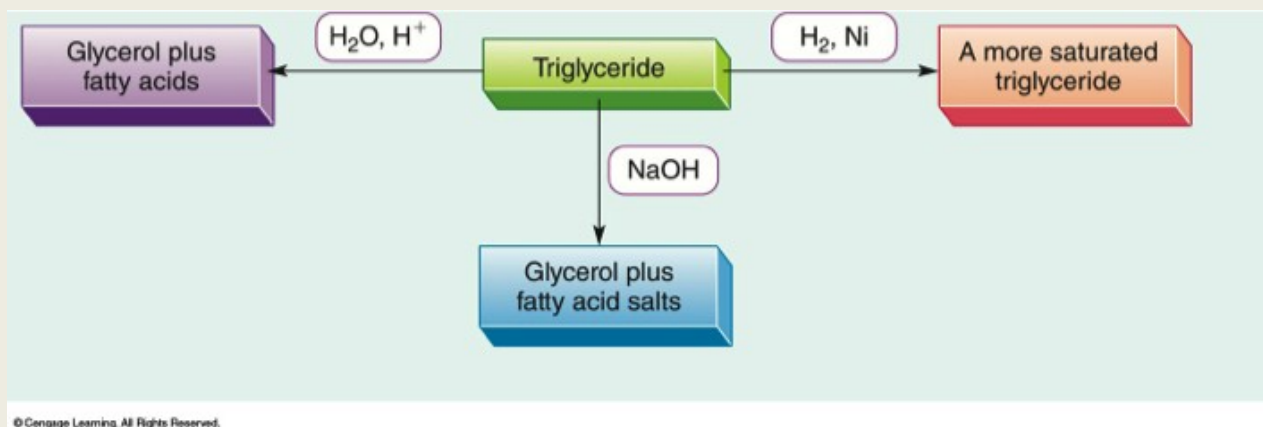
- Fats and oils are triglycerides (triacylglycerols) which are triesters of glycerol



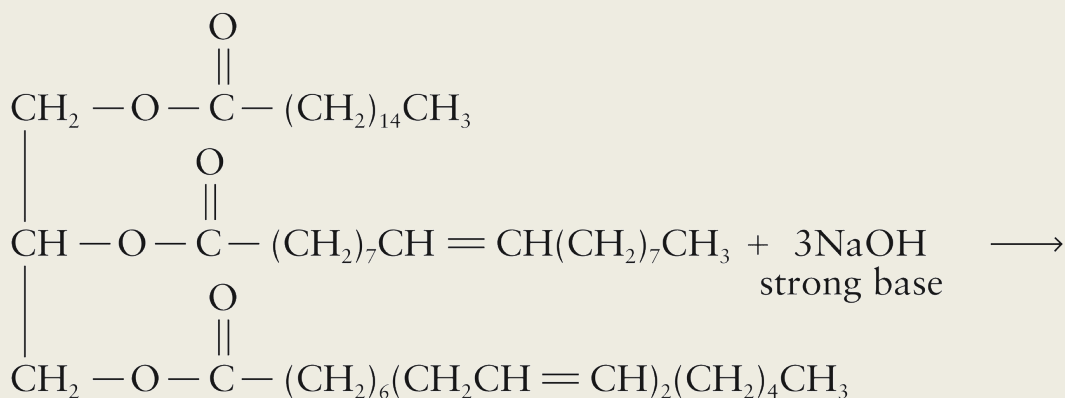
Structure of Triglyceride



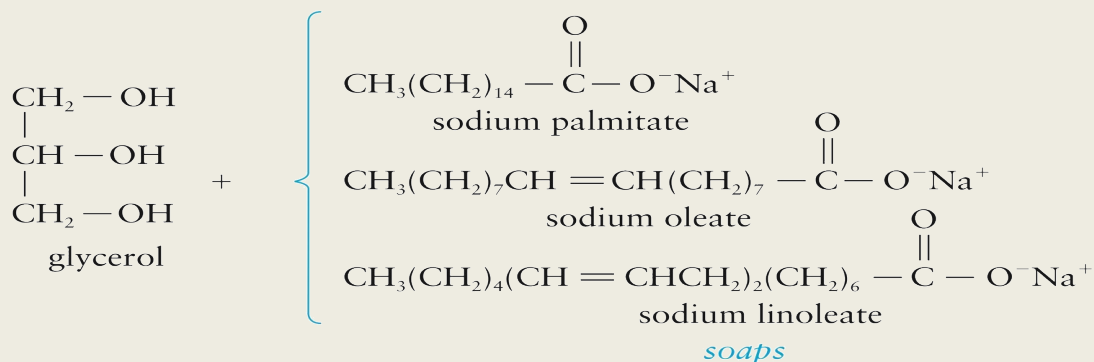
Reactions of Triglycerides



Base Hydrolysis of Triglycerides

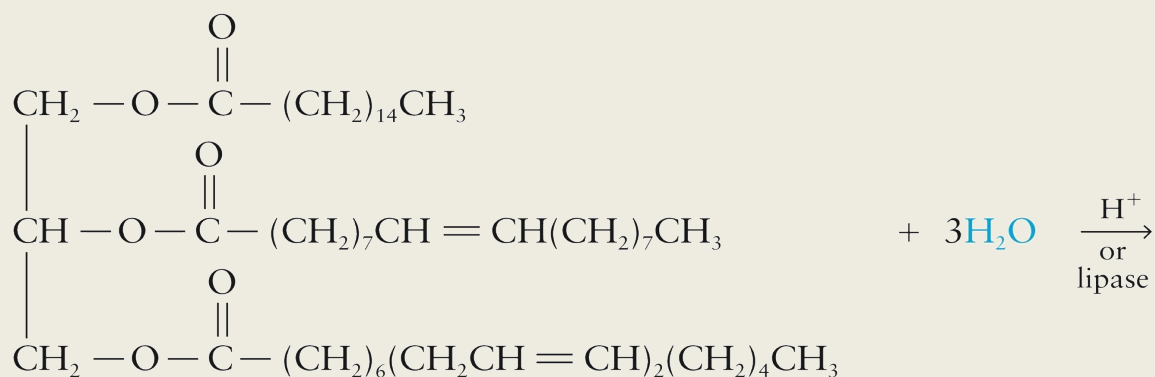


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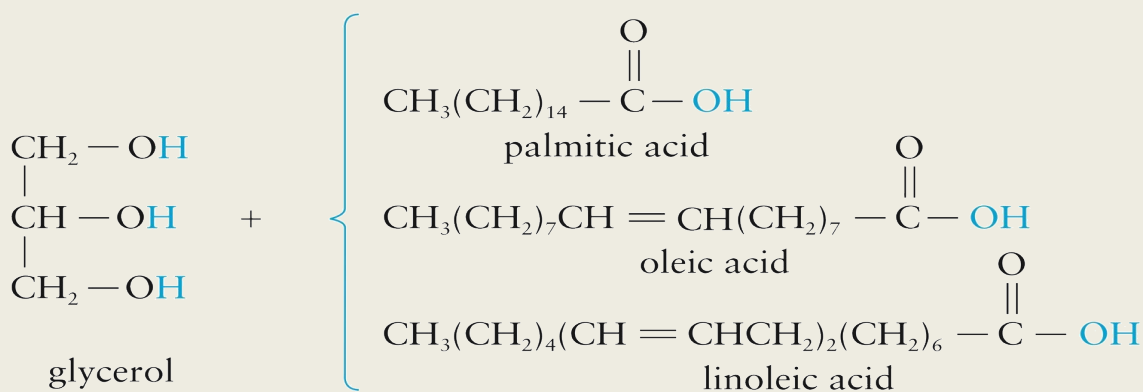


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Acid Hydrolysis of Triglycerides

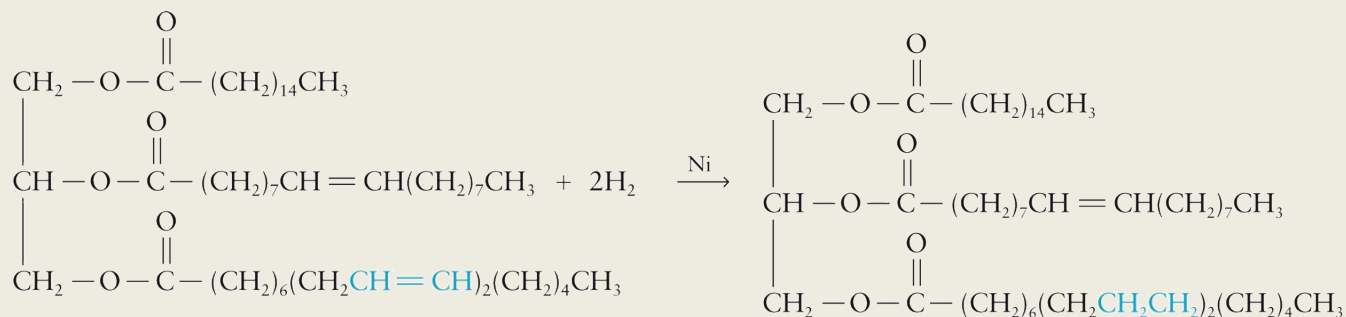


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Hydrogenation of Unsaturated Triglycerides



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double bond has
been reduced to
a single bond

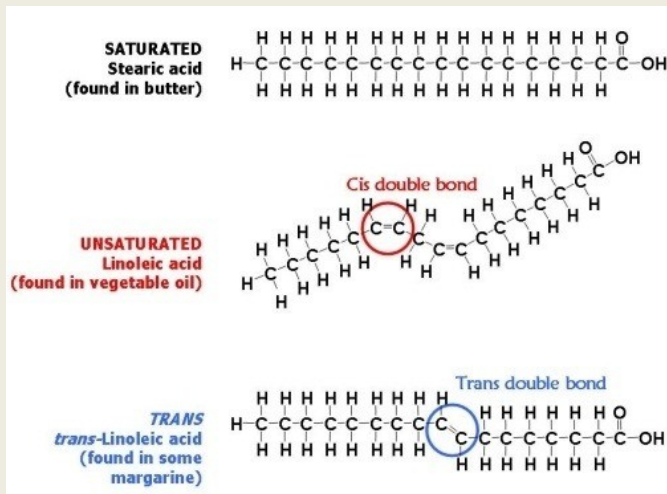
***Trans* Fats**

- Hydrogenation of double bonds results in a decrease in degree of unsaturation
- Increase in melting point of the fat or oil
 - Peanut oil is often partially hydrogenated to convert oil to semisolid that does not separate
 - Hydrogenation is often used in semisolid cooking shortenings such as Crisco and in margarines
- Partially hydrogenated vegetable oils were developed to reduce amount of saturated fats used in frying, baking, and spreads

***Trans* Fats**

- During hydrogenation some fatty acids in *cis* configuration are converted to *trans* configuration
- Recent clinical studies have shown consumption of *trans* fats can raise bad (LDL) cholesterol levels and lower good (HDL) cholesterol levels
- Eating *trans* fats increases risk of developing heart disease and type 2 diabetes

Trans Fats



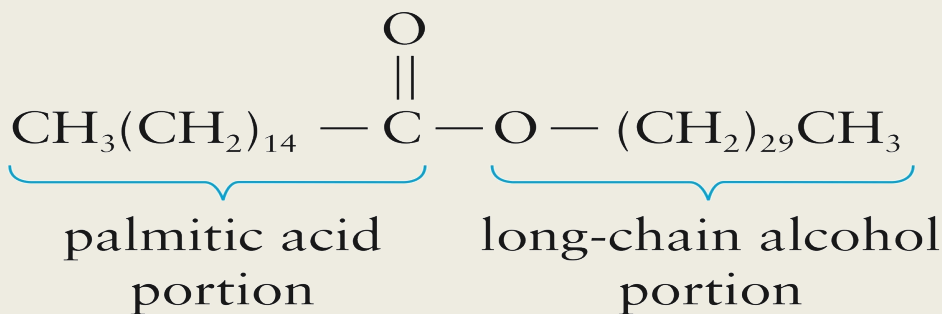
Nutrition Facts		Amount/Serving	%DV*	Amount/Serving	%DV*		
Serv. Size		Total Fat	7g	11%	Total Carb.	20g	7%
4 cookies (32g)		Sat. Fat	4.5g	23%	Dietary Fiber	1g	4%
Serving 9		Trans Fat	0g		Sugars	10g	
Calories 150		Cholest.	0mg	0%	Protein	2g	
Calories from fat 60		Sodium	115mg	5%			
Vitamin A 0% • Vitamin C 0% • Calcium 0% • Iron 4%							

INGREDIENTS: Enriched flour, riboflavin, sugar, partially hydrogenated vegetable oil, cocoa, cornstarch, hydrogenated oils, soy lecithin, salt, caramel color, artificial flavors.

FDA has given industry to 2018
to remove *trans* fats from food

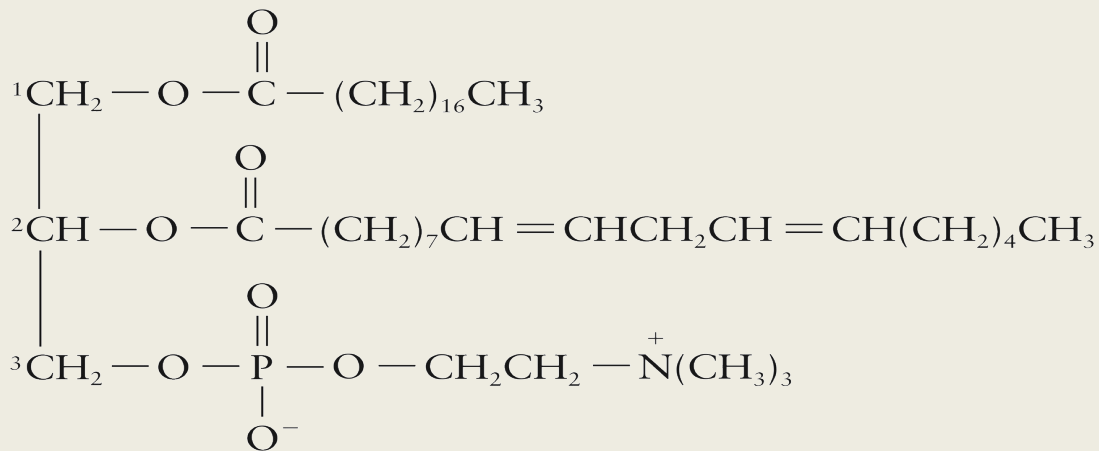
Waxes

- Waxes are esters of fatty acids and long chain alcohols (12-32 carbon atoms). They are:
 - Water-insoluble and not easily hydrolyzed.
 - Often found in protective coatings on feathers, fur, skin leaves, and fruits.
 - Used commercially to make cosmetics, candles, ointments, and protective polishes.



Phosphoglycerides

- Phospholipid that contains glycerol, fatty acid, phosphoric acid, and an alcohol (usually an amino alcohol)



phosphatidylcholine

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Phosphoglycerides

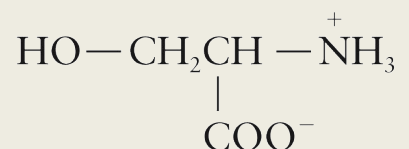
- Complex lipids that serve as the major components of cell membranes
- Also known as phospholipids
- Most abundant phosphoglycerides contain an amino alcohol such as choline, ethanolamine, or serine



choline
(a quaternary ammonium cation)



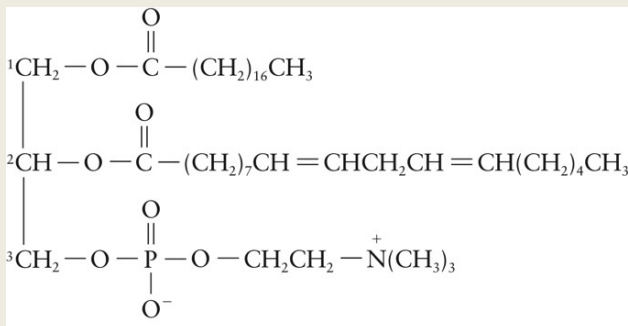
ethanolamine
(cation form)



serine
(two ionic groups present)

Phosphoglycerides

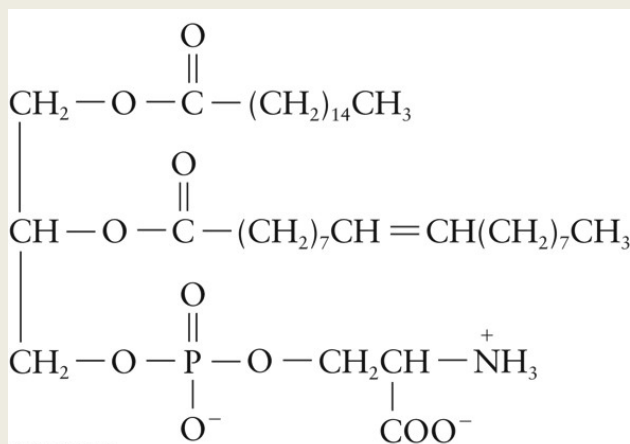
- Lecithins:
 - Contain the amino alcohol choline.
 - Are an important cell membrane component.
 - Are emulsifying and micelle-forming agents.
 - Commercially extracted from soybeans and used to give products such as margarine and chocolate candies a smooth texture



Lecithins
(phosphatidylcholine)

Phosphoglycerides

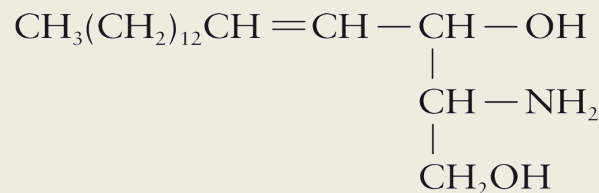
- Cephalins:
 - Contain ethanolamine or serine as the alcohol.
 - Are found in most cell membranes, especially brain tissue and blood platelets (role in blood-clotting process).



Cephalins
(phosphatidylserine)

Sphingolipids

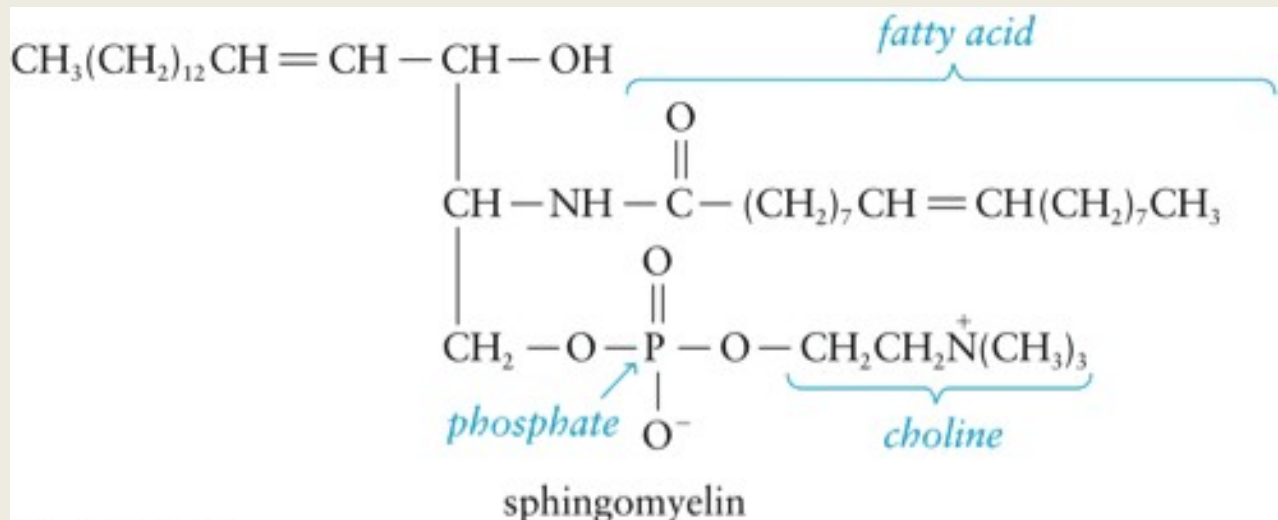
- Sphingolipids are complex lipids that contain a long-chain unsaturated amino alcohol called sphingosine, rather than glycerol.
- Two types of sphingolipids exist:
 - Sphingomyelin contains a phosphate and choline group. It is found in the myelin sheath surrounding nerve cells.
 - Glycolipids contain a carbohydrate unit. They are often called cerebrosides because of their abundance in brain tissue.



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sphingosine

Sphingolipids - Sphingomyelin



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Sphingolipids

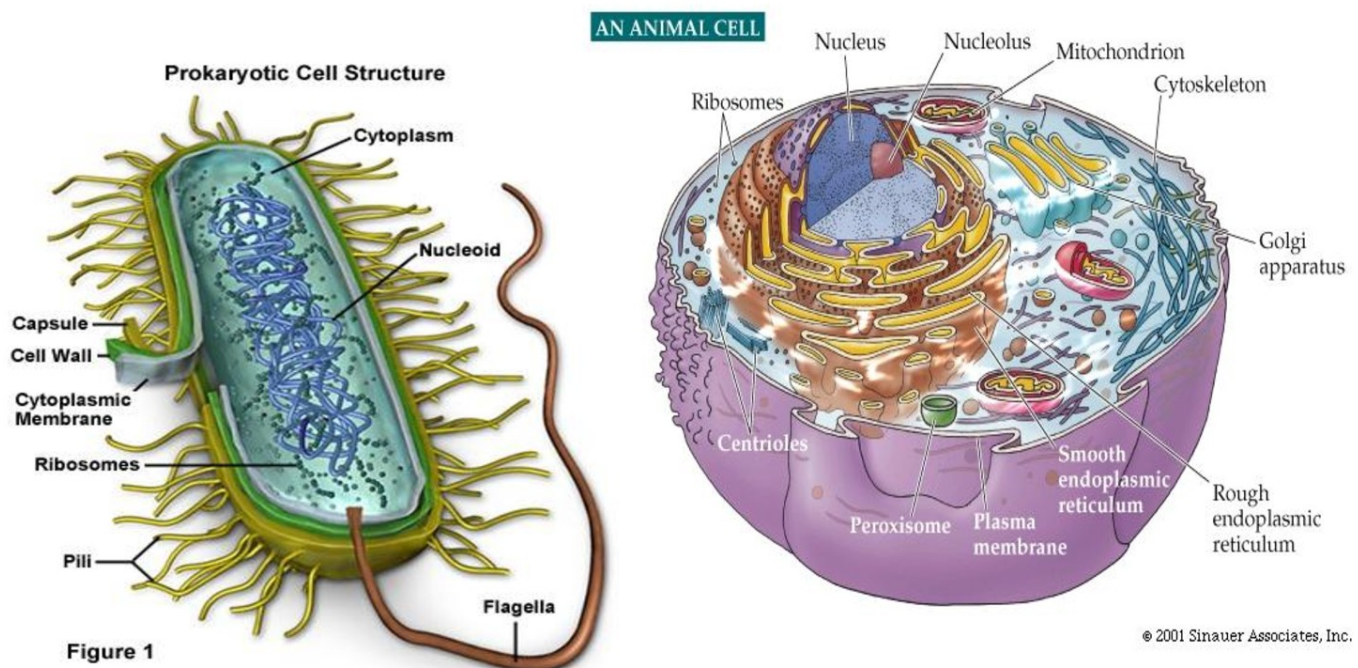
- Several human diseases are known to result from an abnormal accumulation of sphingomyelins and glycolipids in the body.
- Each of these diseases is the result of an inherited absence of an enzyme needed to break down these complex lipids.

TABLE 18.2 Diseases Originating from Abnormal Metabolism and Accumulation of Glycolipids and Sphingomyelins

Disease	Organs(s) Affected	Type of Lipid Accumulated
Tay-Sachs	Brain	Glycolipid (ganglioside G _{M2})
Gaucher's	Spleen, liver	Cerebrosides containing glucose
Niemann-Pick	Several, particularly liver and spleen	Sphingomyelins

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Prokaryotic vs Eukaryotic Cells



Cell Membranes

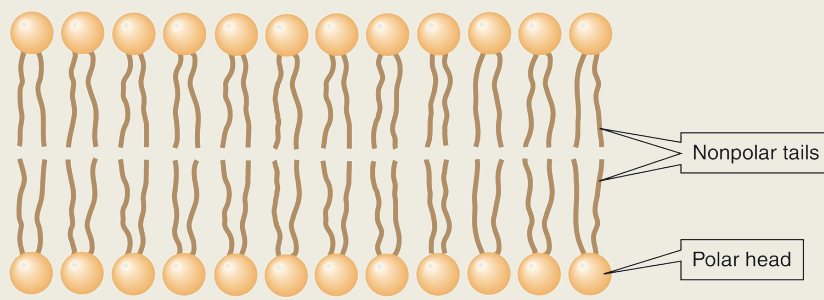
TABLE 18.3 The Functions of Some Cellular Organelles

Organelle	Function
Endoplasmic reticulum	Synthesis of proteins, lipids, and other substances
Lysosome	Digestion of substances taken into cells
Microfilaments and microtubules	Cellular movements
Mitochondrion	Cellular respiration and energy production
Nucleus	Contains hereditary material (DNA), which directs protein synthesis
Plastids	Contains plant pigments such as chlorophyll (photosynthesis)
Ribosome	Protein synthesis

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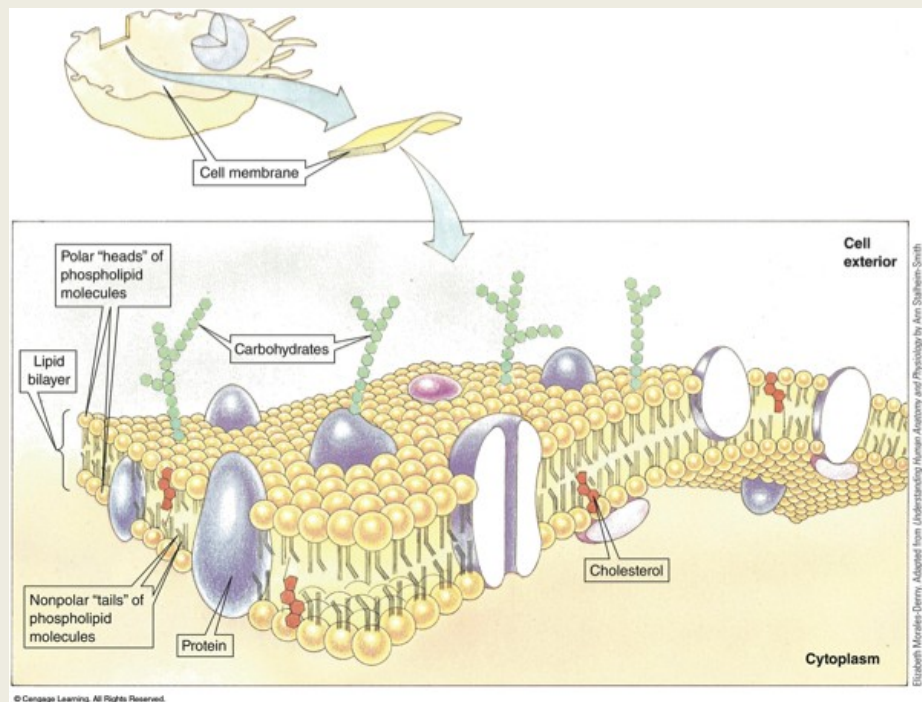
Membrane Structure

- Membranes are organized in a lipid bilayer with hydrophobic (long carbon chain) portions inside and hydrophilic (polar groups) exposed to the water environment.
- Most membranes are 60% lipid and 40% protein.
- The lipids in membranes are phosphoglycerides, sphingomyelin, and cholesterol.
- When a lipid bilayer is broken and the tails are exposed to water, the resulting repulsion causes the bilayer to reform and the break seals spontaneously.

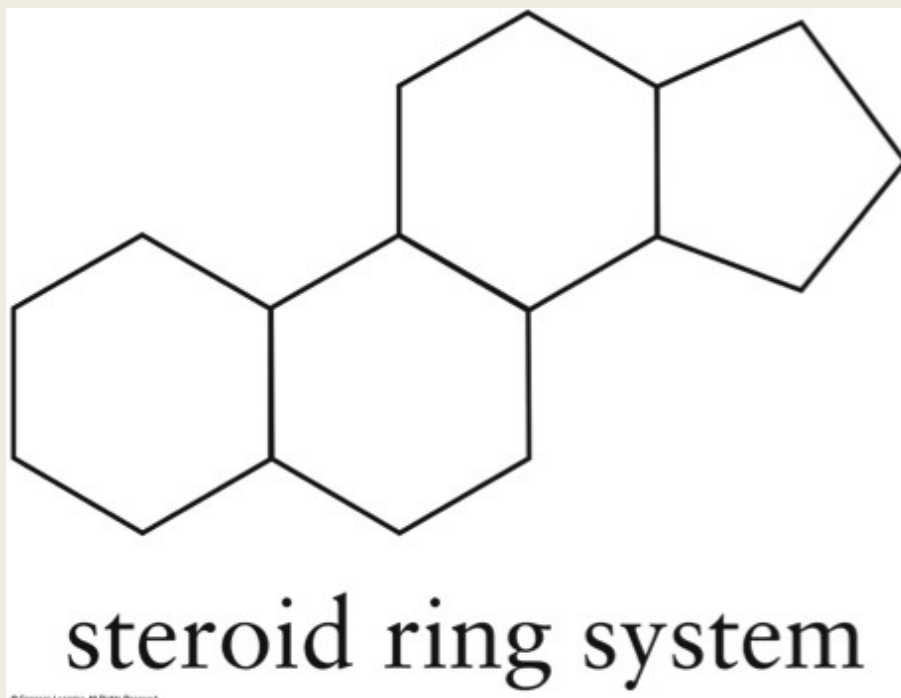


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Fluid-Mosaic Model

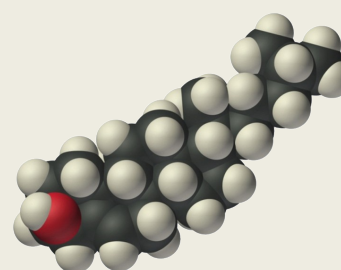
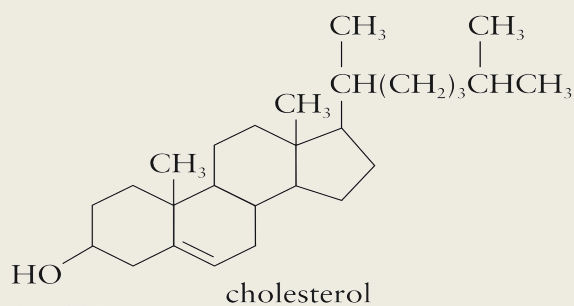


Steroids



Steroids - Cholesterol

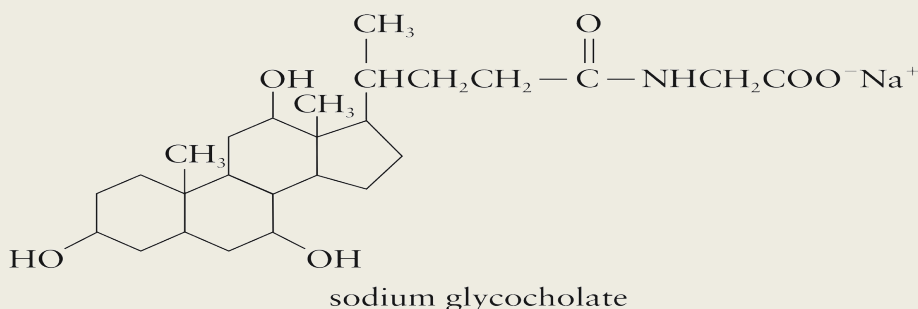
- Most abundant and important steroid in the human body.
- Essential component of cell membranes.
- Precursor of other important steroids.
- Synthesized by liver and present in foods.
- Strong correlation exists between cholesterol blood levels and atherosclerosis (hardening of arteries).



Cholesterol

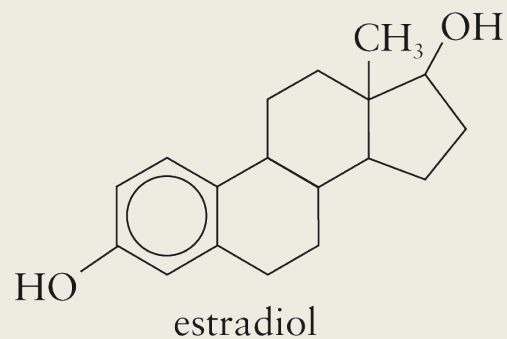
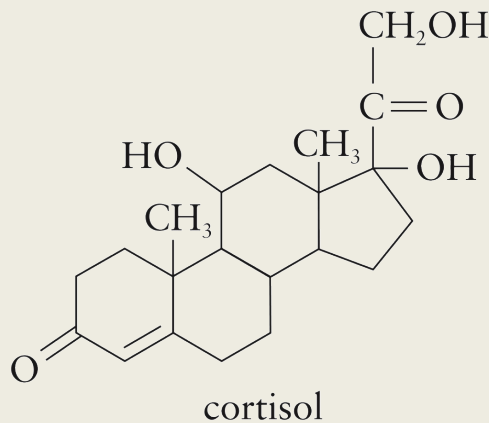
Steroids – Bile Salts

- Bile is yellowish-brown or green, produced by the liver, stored in the gallbladder, and emptied into the intestine.
 - Bile salts emulsify lipids in the intestine by breaking apart lipid globules.
 - The increased surface area allows lipids to be more easily hydrolyzed.
 - Bile salts emulsify cholesterol found in the bile.



Steroid Hormones

- A hormone is a chemical messenger secreted by specific glands and carried through the blood to a target tissue, where it triggers a particular response.
- There are two major categories of steroid hormones:
 - adrenocorticoid hormones
 - sex hormones.

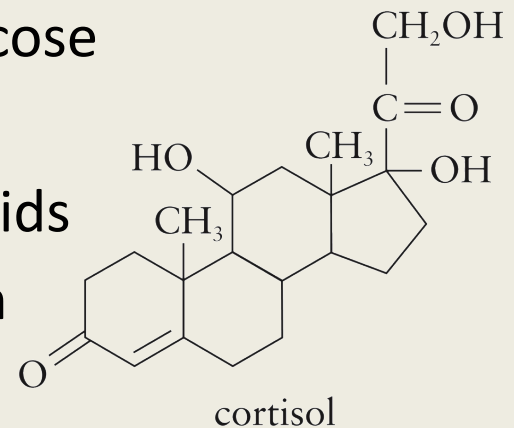


Adrenocorticoid Hormones

- Adrenocorticoid hormones are produced by the adrenal glands.
- Two classes of adrenocorticoid hormones are:
 - Mineralocorticoids regulate the concentration of ions in bodily fluids.
 - Example: Aldosterone
 - Increases absorption of Na^+ and Cl^-
 - Glucocorticoids, enhance carbohydrate metabolism.
 - Example: Cortisol
 - Increases glucose and glycogen concentrations in the body
 - Powerful anti-inflammatory effects in the body

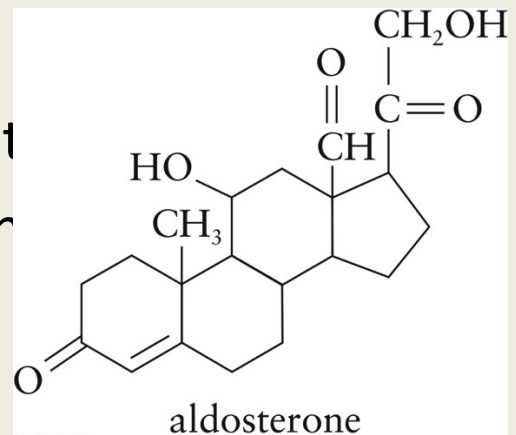
Glucocorticoid - Cortisol

- Most important human glucocorticoid
- Stimulates the expression of anti-inflammatory proteins
- Stimulates the synthesis of glucose from non-carbohydrate sources
- Excessive levels of glucocorticoids interfere with calcium absorption which can lead to osteoporosis



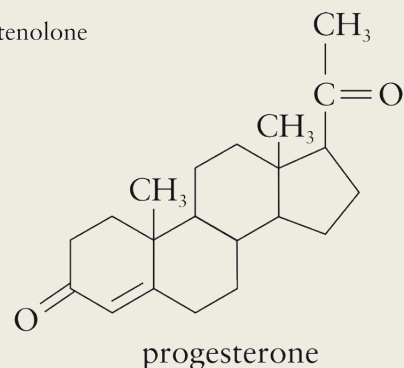
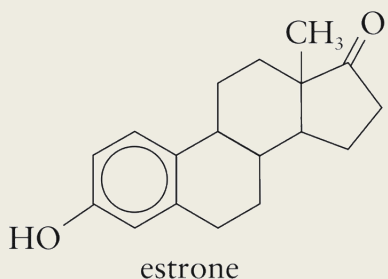
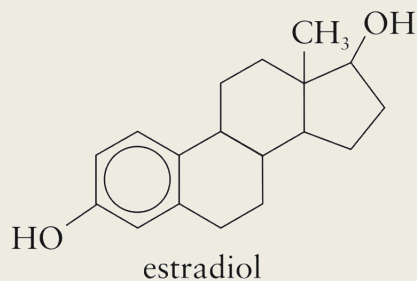
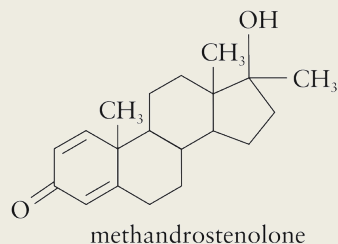
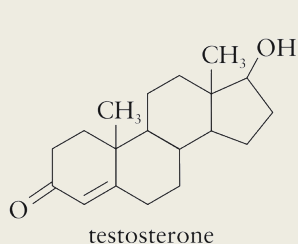
Mineralocorticoid

- Aldosterone is the most important mineralocorticoid.
- An increase results in the increased absorption of sodium and chloride ion.
- The increased absorption of sodium and chloride leads to an increase in the retention of water.



Sex Hormones

- Androgens are male sex hormones produced by testes, including testosterone.
- Estrogen and progesterone are female sex hormones produced by ovaries, including estradiol and estrone (the most important estrogen).



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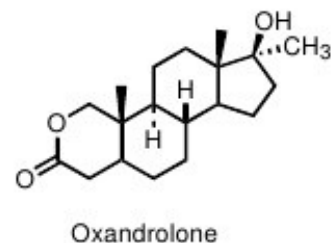
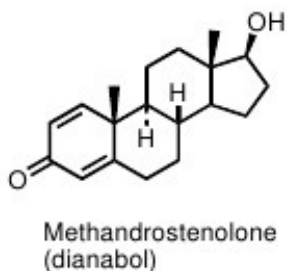
Anabolic Steroids



Lacks methyl group



Ester as prodrug

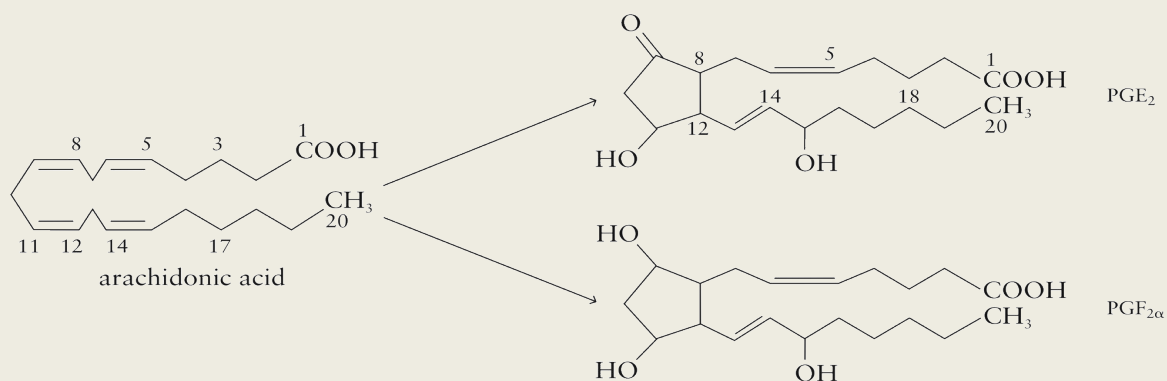


Anabolic Steroids

- Increase rate of protein synthesis in cells, especially skeletal muscle
- Induce development of masculine secondary sexual characteristics such as facial and body hair
- While anabolic steroids have legitimate medical uses, they are often taken by athletes as performance-enhancing drugs (PED)
- Side effects include
 - hypertension (high blood pressure)
 - decreased high-density lipoproteins (HDL-“good cholesterol”); increased low-density lipoprotein (LDL-“bad cholesterol”)
 - psychiatric disorders such as depression and aggressive behavior and rage (“roid rage”)

Prostaglandins

- Cyclical compounds synthesized from 20-carbon unsaturated fatty acid, arachidonic acid.
- Similar to hormones in that they are intimately involved in a host of body processes.
 - Involved in controlling inflammation and fever
 - Involved in almost every phase of reproduction.
 - Involved in blood clotting.



How does aspirin work in people?

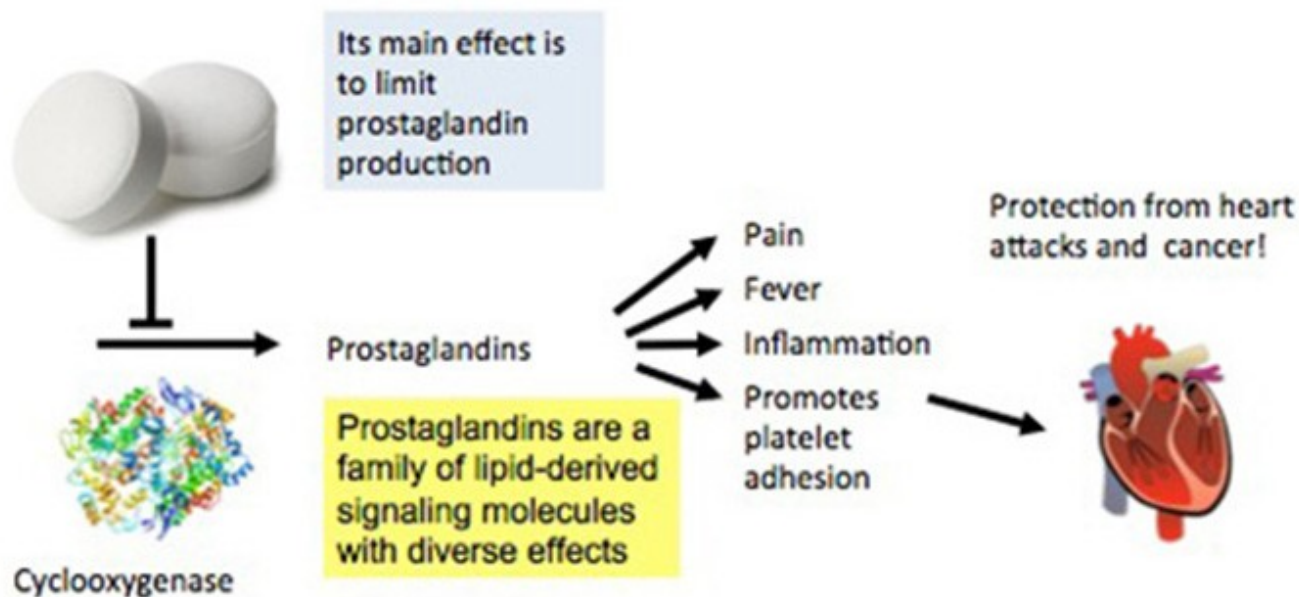
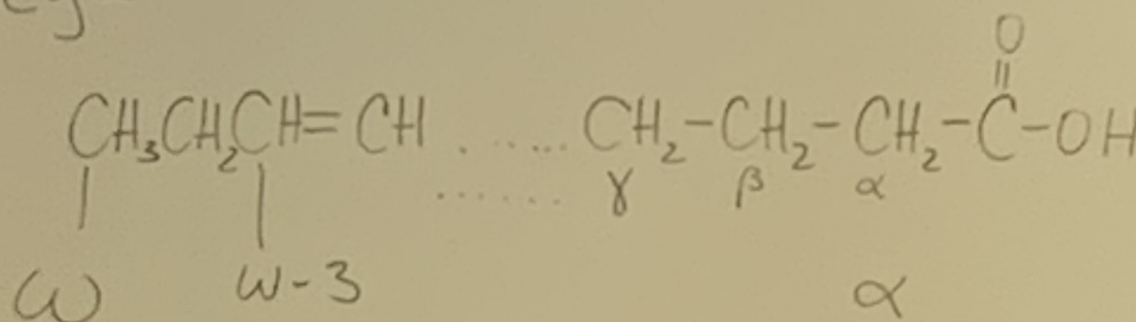


Image courtesy of [aspirin.com](#)

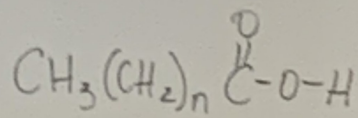
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omega

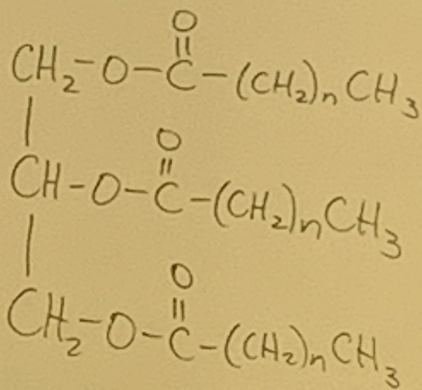
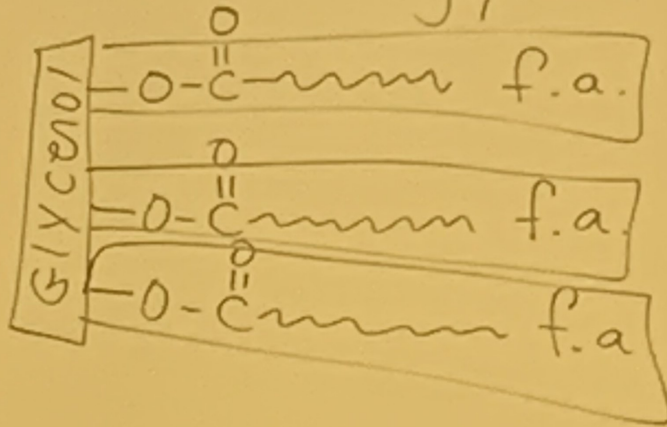


Omega-3 C=C 3 C's from end
 omega-6 C=C 6 " " "

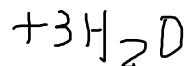
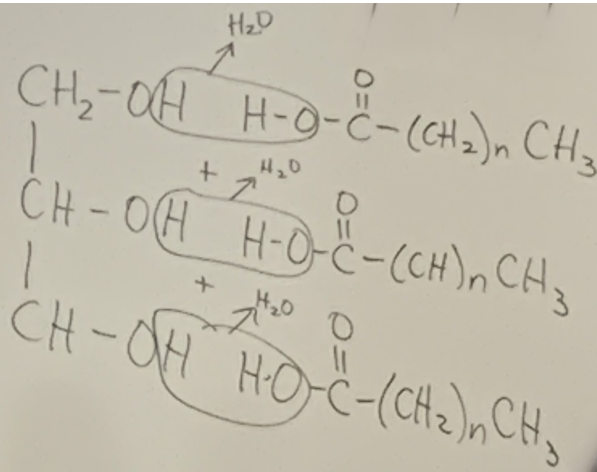
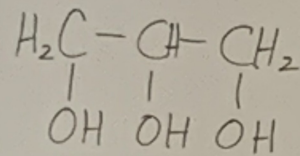
triglyceride

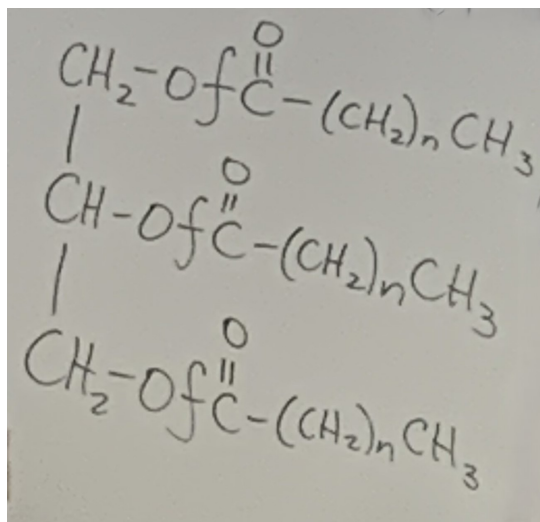


triester of glycerol + 3 f.a.

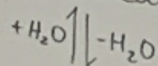


glycerol-trialcohol





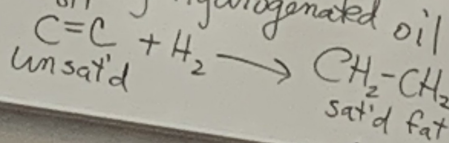
① Esterification



② hydrolysis

③ hydrogenation

partially hydrogenated oil



semisoft fat